

Lab 5: Verification of Superposition Theorem.

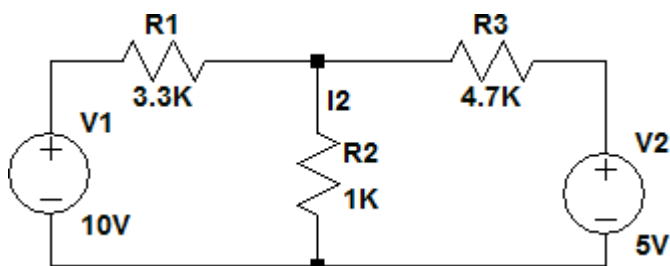
Objective:

- To verify Superposition Theorem.

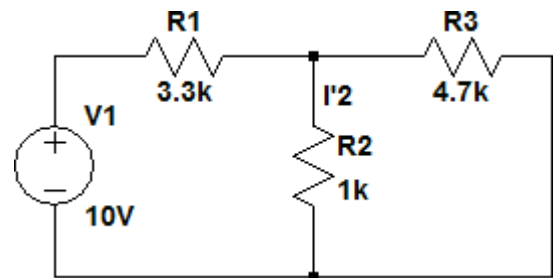
List of Equipment

- Breadboard
- DMM
- 1 x $3.3k\Omega$ resistor
- 1 x $4.7k\Omega$ resistor
- 1 x $1K\Omega$ resistor

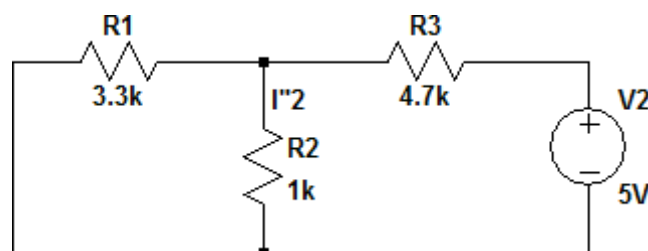
Circuit Diagram



Circuit 1



Circuit 2



Circuit 3

Procedure:

1. Set up Circuit 1.
2. Mark the polarities of each resistor.
3. With both the voltage source connected to the circuit, measure I_2 , V_{R1} , V_{R2} , V_{R3} and record the values in appropriate tables.
4. Setup Circuit 2. Measure and record I'_2 , V'_{R1} , V'_{R2} , V'_{R3} .
5. Setup Circuit 3. Measure and record I''_2 , V''_{R1} , V''_{R2} , V'_{R3} .



Data Collection for Lab5:

Group No. _____

Instructor's Signature _____

Table 1:

From Circuit 1	From Circuit 2	From Circuit 3	
I_{R2}	I'_{R2}	I''_{R2}	$I'_{R2} + I''_{R2}$

Table 2:

From Circuit 1	From Circuit 2	From Circuit 3	
V_{R1}	V'_{R1}	V''_{R1}	$V'_{R1} + V''_{R2}$

Table 3:

From Circuit 1	From Circuit 2	From Circuit 3	
V_{R2}	V'_{R2}	V''_{R2}	$V'_{R2} + V''_{R2}$

Table 4:

From Circuit 1	From Circuit 2	From Circuit 3	
V_{R3}	V'_{R3}	V''_{R3}	$V'_{R3} + V''_{R3}$

Verification of Superposition Theorem:

➤ Is $I_{R2} = I'_{R2} + I''_{R2}$? Answer: _____

➤ Is $V_{R1} = V'_{R1} + V''_{R1}$? Answer: _____

➤ Is $V_{R2} = V'_{R2} + V''_{R2}$? Answer: _____

➤ Is $V_{R3} = V'_{R3} + V''_{R3}$? Answer: _____



Report:

1. What is Superposition Theorem?
2. Theoretically calculate all values of Table 1 to Table 4. **Show all the steps in details.**
3. Using measured data, show that your circuit followed superposition theorem.
4. Find the % Error between your theoretical and experimental values.